

Context Free Language / Grammar

[Chapter 4 in John C Martin]

Context Free Grammar

A context free grammar (CFG) is a 4-tuple $G = (V, T, S, P)$ where V and T are disjoint finite sets,

$S \in V$,

P is a finite set of formulae/productions of the form $A \rightarrow \alpha$ where $A \in V$ and $\alpha \in (V \cup T)^*$

Elements of T are called terminal symbols

Elements of V are called variables (non-terminals)

S is the start symbol

Elements of P are grammar rules or productions

The Language Generated by CFG

If $G = (V, T, S, P)$ is a CFG, the language generated by G is

$$L(G) = \{ \alpha \in \Sigma^* \mid S \xrightarrow{*}_G \alpha \}$$

A language L is a context free language (CFL) if there is a CFG G with $L = L(G)$.

Q. Write a grammar and its rules to generate even length strings.

Ans. $L = \{ \alpha \mid |\alpha| \text{ is even} \}$

Valid = $\{ \epsilon, aa, ab, ba, bb, \dots \}$

$\Sigma = \{a, b\}$

$V = \{S\}$

$S = S$

$P = \{ S \rightarrow \epsilon$

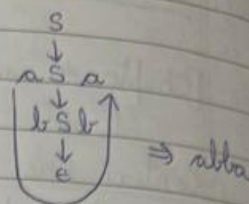
$S \rightarrow aSa \mid aSb \mid bSa \mid bSb$

$\}$

→ ANSWER-1

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Eg: abba



Rules:

- 1) ϵ belongs to language
- 2) $ab \Rightarrow aSa, aSb, bSa, bSb$

$L = \{\epsilon, aa, ab, ba, bb, \dots\}$

$S \rightarrow \epsilon$

$S \rightarrow Saa | Sab | Sba | Sbb$

→ ANSWER-2

$S \rightarrow \epsilon$

$S \rightarrow aaS | abS | baS | bbS$

→ ANSWER-3

$S \rightarrow \epsilon$

$S \rightarrow aS_1 | bS_1$

$\Rightarrow$

$S_1 \rightarrow aS | a$

$S_1 \rightarrow bS | b$

ANSWER-4

~~$S \rightarrow \epsilon$~~

~~$S \rightarrow aS_1 | bS_1$~~

~~$S_1 \rightarrow aS | \epsilon$~~

~~$S_1 \rightarrow bS | \epsilon$~~

ODD LENGTH

$S \rightarrow aS_1 | bS_1 | \epsilon$

$S_1 \rightarrow aS | bS$

ANSWER-5

~~$S \rightarrow aS_1 | bS_1 | \epsilon$~~

~~$S_1 \rightarrow aS | bS | \epsilon$~~

ODD LENGTH

Q. Write a CFG to generate the strings of the language
 $L = \{w \mid |w| \text{ is odd and middle symbol is 'a'}\}$

Ans. $L = \{a, aaa, aab, baa, bab, \dots\}$

Rules:

1) $a \in L$

2)

$P = \{ S \rightarrow a$

$S \rightarrow aSa | aSb | bSa | bSb \}$

$V = \{S\}$

$S = S$

$\Sigma = \{a, b\}$

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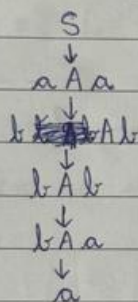
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Q. Write a CFG to generate the strings of the language:

 $L = \{w \mid \text{first and middle symbol are same, } |w| \text{ is odd}\}$ Ans. $L = \{aab, bba, aaa, bbb, \dots\} \rightarrow a, b \notin L$ as

First & middle elements are not differentiable

 $V = \{S, A, B\}$ $T = \{a, b\}$ $S = S$ $P = \{$ $S \rightarrow aAb \mid aAa \mid bBa \mid bBb$ $A \rightarrow a \mid aAa \mid aAb \mid bAa \mid bAb$ $B \rightarrow b \mid aBa \mid aBb \mid bBa \mid bBb$ $\}$ $S \rightarrow aAb \mid aAa \mid bBa \mid bBb$ $A \rightarrow a \mid aAa \mid aAb \mid bAa \mid bAb$ $B \rightarrow b \mid aBa \mid aBb \mid bBa \mid bBb$ Definition Tree for $x = abbaabba \in L$:

$$\begin{array}{c}
 abbaabba \\
 bbaab \\
 bbaab \\
 ba \\
 a
 \end{array}$$

Q. Write a CFG to generate the strings of the language

 $L = \{w \mid \text{first & last symbols are same, } |w| \text{ is even}\}$ Ans. $L = \{\epsilon, aa, bb, abaa, abba, baab, baab, \dots\}$ $V = \{S, A, B\}$ $T = \{a, b\}$ $S = S$ $P = \{ S \rightarrow \epsilon \mid aS, a \mid bS, b$ $S_1 \rightarrow aS_1, a \mid aS_1, b \mid bS_1, a \mid bS_1, b \mid \epsilon \}$

Q. Write a CFG to generate the strings of the language

 $L = \{w \mid |w| \text{ is odd, middle & last symbol are same}\}$ Ans. $P = \{ S \rightarrow aAa \mid aBb \mid bAa \mid bBb$ $A \rightarrow a \mid aAa \mid aAb \mid bAa \mid bAb$ $B \rightarrow b \mid aBa \mid aBb \mid bBa \mid bBb \}$

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Q. $L = \{w \mid |w| \text{ is odd, first, middle \& last symbols are same}\}$
 Ans. $S \rightarrow aAa \mid bBb$
 $A \rightarrow a \mid aAa \mid aAb \mid bAa \mid bAb$
 $B \rightarrow b \mid bBb \mid aBb \mid bBa \mid aBa$

Q. $L = \{a^n b^n \mid n \geq 0\}$
 Ans. $L = \{ab, aabb, \dots\}$
 $S \rightarrow ab$
 $S \rightarrow aSb$

$L = \{a^n b^n \mid n \geq 0\}$
 $L = \{e, ab, aabb, \dots\}$
 $S \rightarrow e$
 $S \rightarrow aSb$

Q. $L = \{a^n b^n \mid n \geq 2\}$
 Ans. $S \rightarrow aabb$
 $S \rightarrow aSb$

Q. $L = \{w \mid ww^R\}$
 Ans. $L = \{e, aa, bb, abba, baab, \dots\}$
 $S \rightarrow e$
 $S \rightarrow aSa \mid bSb$

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Q. Write a CFG for the language $L = \{w \mid |w| \text{ is odd}\}$
 Ans. $L = \{a, b, aaa, abba, baab, bbb, abb, aab, baa, bba, \dots\}$
 1) a & b are in the L
 2) If $w \in L$, so is awa, awb, bwa, bwb are in L
 $S \rightarrow a \mid b$
 $S \rightarrow aSa \mid aSb \mid bSa \mid bSb$

1. CFG for the language $L = \{w \mid |w| \text{ is odd}\}$ Rules
 1) a, b are in the language.
 2) If $w \in L$ then so is awa, awb, bwa, bwb are in the language and others are not.

2. CFG for the CFL $L = \{a^n b^n \mid n \geq 0\}$
 RULES
 1) e belongs to the language.

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2) If $w \in L$, then so is $awb \in L$ and others don't.

PRODUCTIONS

 $S \rightarrow \epsilon$ $S \rightarrow aSb$

3. CFG for the CFL $L = \{a^n b^n \mid n \geq 0\}$

RULES

1) $ab \in L$.2) If $w \in L$, then so is $awb \in L$ and others don't.

PRODUCTIONS

 $S \rightarrow ab$ $S \rightarrow aSb$

4. CFG for CFL $L = \{a^n b^n \mid n \geq 2\}$

RULES

1) $aabb \in L$.2) If $w \in L$, then so is $awb \in L$ and others don't.

PRODUCTIONS

 $S \rightarrow aabb$ $S \rightarrow aSb$

Q. Write CFG for the language $L = \{a^m b^m c^k \mid m, m, k \geq 0, n = m+k\}$

Ans.

$$a^{m+k} b^m c^k$$

$$\Rightarrow a^m a^k b^m c^k$$

$$\Rightarrow a^k a^m b^m c^k$$

$$\text{Valid} = \{\epsilon, ab, ac, aabc, aabb, aacc, \dots\}$$

$$S \rightarrow aSc \mid \epsilon \mid S_1$$

$$S_1 \rightarrow aS_1 b \mid ab$$

Q. Generate CFG for the language $L = \{a^n b^m \mid n \geq m\}$

Ans.

$$L = \{\epsilon, a, ab, a^2, a^2a, a^2ab, \dots\}$$

$$S \rightarrow \epsilon$$

$$S \rightarrow aSb \mid A$$

$$A \rightarrow aA \mid a$$

$$\text{Let } k \geq 0, n = m+k$$

$$a^n b^m = a^{m+k} b^m = a^k a^m b^m$$

$$= a^k a^m b^m$$

NOTE: For $n > m$, remove $S \rightarrow \epsilon$. Rest are same.

P.T.O.

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Q. CFG for $L = \{a^i b^j c^k \mid i, j, k \geq 0, j=i \text{ or } j=k\}$

$S_1 \xrightarrow{j=i} a^i b^i c^k$

$L = \{ \epsilon, abc, ab, c, \dots \}$

$S_1 \rightarrow aS_2 b \mid c$

$c \rightarrow cc \mid \epsilon$

$S_2 \rightarrow aS_2 b \mid \epsilon$

$S_3 \rightarrow S_1 \mid S_2$

$S_3 \rightarrow A b S_4 c$

$A \rightarrow aA \mid \epsilon$

$S_4 \rightarrow bS_4 b \mid \epsilon$

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Left Most Derivation and Right Most Derivation

A derivation in CFG is a left most derivation if at each step a product is applied to the left variable occurrence in the current string. A derivation in the CFG is a RMD if at each step a product is applied to the right most variable occurrence in the current string.

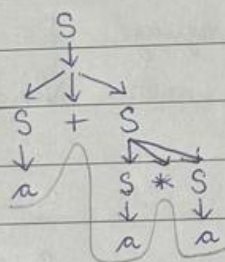
Consider the grammar $S \rightarrow S+S \mid S*S \mid (S) \mid a$

$x = a + a * a$

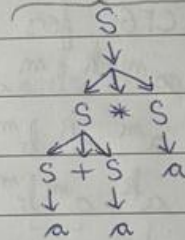
$x \in L(G)$

Parse Tree:

(Both are LMDs)



Does not follow
BODMAS rule so
avoid



Derivation: $S \Rightarrow S+S \Rightarrow a+S*S \Rightarrow a+a*S \Rightarrow a+a*a$

If G has more than 1 LMD/RMD, then G is ambiguous

$G_1: S \rightarrow S+S \mid S*S \mid a$

$S \rightarrow S+T \mid T$

$T \rightarrow T * F \mid F$

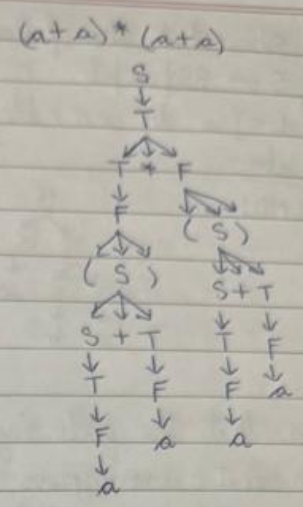
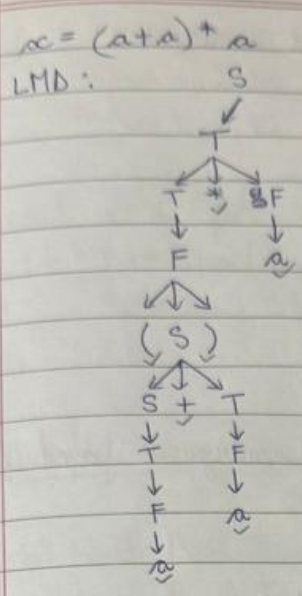
$F \rightarrow (S) \mid a$

$T \rightarrow \text{Term}$

$F \rightarrow \text{Factors}$

$L(G) = \{ a + (a * a), (a + a) * a, a * a * a, \dots \}$

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Ambiguity in CFG

A CFG (G) is ambiguous if for atleast one $x \in L(G)$, x has more than one derivation tree or equivalently more than one LMD or equivalently more than one RMD.

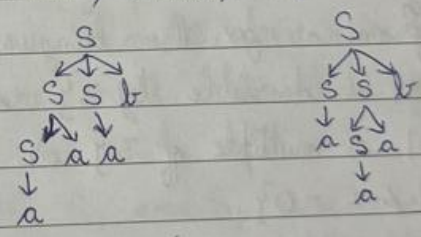
Q. Show that the following grammar is ambiguous:

$$S \rightarrow a | Sa | bSS | SSb | SbS$$

Ans. $L(G) = \{a, aa, aaa, aba, aab, \dots\}$

aaab:

LMD:

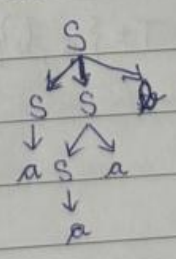


Derivations:

$$S \Rightarrow SSb \Rightarrow aSb \Rightarrow aSabb \Rightarrow aaab$$

$$S \Rightarrow SSb \Rightarrow SaSb \Rightarrow aSaSb \Rightarrow aaab$$

RMD:



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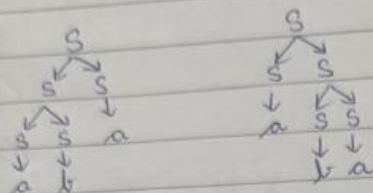
Q. Show that the following grammar is ambiguous:

$$S \rightarrow SS | a | b$$

Ans. $L = \{a, b, aa, bb, aba, \dots\}$

aba:

LMD:



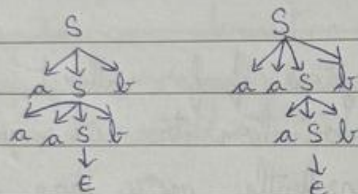
Q. Show that the following grammar is ambiguous and find an equivalent unambiguous grammar:

$$S \rightarrow aSb | aaSb | e$$

Ans. $L = \{e, ab, aab, aaab, aabb, \dots\}$

aaab:

LMD:



$\therefore$ Grammar is ambiguous

Equivalent unambiguous grammar:

$$S \rightarrow aSb | S_1$$

or

$$S \rightarrow aaSb | S_1$$

$$S_1 \rightarrow aaS_1b | e$$

$$S_1 \rightarrow aS_1b | e$$

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Q. Write a CFG for the language over $\Sigma = \{a, b\}$:

$L = \{w \mid |w| \text{ is divisible by } 3\}$ or

$\{w \mid |w| \text{ is multiple of } 3\}$ or

$$\{w \mid |w| \% 3 = 0\}$$

Ans. $L = \{e, aaa, lbb, abba, baab, aabb, bbaa, abbb, \dots\}$

$P = \{$

$$S \rightarrow aS_1 | bS_1 | e$$

$$V = \{S, S_1, S_2\}$$

$$S_1 \rightarrow aS_2 | bS_2$$

$$S = S$$

$$S_2 \rightarrow aS | bS$$

$$T = \{a, b\}$$

$\}$

Q. Write CFG for language over $\Sigma = \{a, b\}$:

$$L = \{w \mid |w| \% 3 \neq 0\}$$

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Ans. $P = \{ S \rightarrow S_1 a \mid S_1 b$
 $S_1 \rightarrow S_2 a \mid S_2 b \mid \epsilon$
 $S_2 \rightarrow S a \mid S b \mid \epsilon \}$

$V = \{ S, S_1, S_2 \}$
 $T = \{ a, b \}$
 $S = S$

Q. Write CFG for $L = \{ w \mid |w| \neq 2 = 0 \}$

Ans. $P = \{ S \rightarrow S_1 a \mid S_1 b \mid \epsilon$
 $S_1 \rightarrow S a \mid S b \}$

$V = \{ S, S_1 \}$
 $T = \{ a, b \}$
 $S = S$

Simplification of Grammar

- S1) Null Production Removal
- S2) Unit Production Removal
- S3) Useless Production / Variable Removal
- S4) Left Recursion Removal

Normal Forms

i) Chomsky Normal Form (CNF)

A context free grammar is said to be in Chomsky Normal Form if every production is one of these 2 types:

- $A \rightarrow BC$ where B and C are variables
- $A \rightarrow \alpha$ where α is a terminal symbol

→ To arrive at CNF, S_1 and S_2 are mandatory.

→ S_3 is not mandatory, but is preferable as it reduces number of productions

Removal of Null Production

A production of the form $A \rightarrow \epsilon$ is known as null production or ϵ production.

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Nullable variable

A variable V , S is said to be nullable variable if it derives ϵ in zero or more steps, i.e., $S^* \Rightarrow \epsilon$.

Q. Find nullable variables from the grammar

G: $S \rightarrow ABA$

$A \rightarrow aA | B$

$B \rightarrow bB | \epsilon$

Ans. $B \Rightarrow \epsilon$

$A \Rightarrow B \Rightarrow \epsilon$ so $A \Rightarrow B \Rightarrow \epsilon$

$S \Rightarrow ABA \Rightarrow BBA \Rightarrow BA \Rightarrow A \Rightarrow B \Rightarrow \epsilon$ so $S \Rightarrow \epsilon$

Nullable variables, $NV = \{S, A, B\}$

Q. Find the NV for the given grammar:

G: $S \rightarrow ABA$

$A \rightarrow aA | a$

$B \rightarrow bB | \epsilon$

Ans. $NV = \{B\}$

To find the equivalent grammar without null-productions, the steps are:

- 1) Identify nullable variables.
- 2) Copy all the productions of G into G' except null productions.
- 3) Remove nullable variables from the body of each production and introduce new production.

Q. $G: S \rightarrow AB | \epsilon$

$A \rightarrow aASb | a$

$B \rightarrow bS$

Remove null-productions for G .

Ans. $NV = \{S\}$

$G': S \rightarrow AB$

$A \rightarrow aASb | a | aAb$

$B \rightarrow bS | b$

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Q. Remove nullable variable for grammar:

G: $S \rightarrow AB$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow ab \mid bB \mid \epsilon$

Ans.

$NV = \{S, A, B\}$

$G': S \rightarrow AB \mid B \mid A$

$A \rightarrow aA \mid a$

$B \rightarrow ab \mid bB \mid b$ ← Start from here

Q. Remove nullable variable from grammar:

G: $S \rightarrow ABC \mid BaB$

$A \rightarrow aA \mid BaC \mid aac$

$B \rightarrow bBb \mid a$

$C \rightarrow CA \mid aAC$

Ans. No nullable productions -1M

Q. Remove NV for:

G: $S \rightarrow AB \mid ABC$

$A \rightarrow BA \mid BC \mid a \mid \epsilon$

$B \rightarrow AC \mid CB \mid b \mid \epsilon$

$C \rightarrow BC \mid AB \mid A \mid a$

Ans. $NV = \{S, A, B, C\}$

$G': S \rightarrow AB \mid ABC \mid B \mid A \mid BC \mid AC \mid C$

$A \rightarrow BA \mid BC \mid a \mid B \mid A \mid C$

$B \rightarrow AC \mid CB \mid b \mid C \mid A \mid B$

$C \rightarrow BC \mid AB \mid A \mid a \mid C \mid B$

Unit Productions

A unit production is of the form $A \rightarrow B$ where A and B are variables.

Unit productions are removed by substitution.

Eg: G: $S \rightarrow aS \mid B \mid bB$

$B \rightarrow bB \mid b$

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After removing unit productions:

$$G': S \rightarrow aS \mid bB \mid b$$

$$B \rightarrow bB \mid b$$

NOTE: $S \rightarrow b$ is not unit production as RHS is a terminal while
 $S \rightarrow B$ is unit production as RHS is a variable.

Q. Remove null production and unit production for:

$$G: S \rightarrow AB$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

Ans. $NV = \{S, A, B\}$

Step 1: Nullable Variables Removal

$$S \rightarrow AB \mid A \mid B$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

Step 2: Unit Productions Removal

$$S \rightarrow AB \mid aA \mid a \mid bB \mid b$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

Q. Convert the above grammar to CNF.

Ans. $G: S \rightarrow AB \mid a \mid b \mid X_a A \mid X_b B$

$$X_a \rightarrow \#a$$

$$X_b \rightarrow \#b$$

$$A \rightarrow X_a A \mid a$$

$$B \rightarrow X_b B \mid b$$

Q. Find equivalent CFG with no useless variables/productions:

$$G: S \rightarrow ABC \mid BaB\#$$

$$A \rightarrow aA \mid BaC \mid aca$$

$$B \rightarrow bBb \mid a$$

$$C \rightarrow CA \mid AC$$

Ans. C is the useless variable as it is non-generating even though it is reachable.

non generating:
does not produce any terminals

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Hence, the equivalent CFG is

$$S \rightarrow BaB$$

$$A \rightarrow aA / a$$

$$B \rightarrow bBb / a$$

Now A is not reachable, so A is useless variable.
Hence, the equivalent CFG is

$$S \rightarrow BaB$$

$$B \rightarrow bBb / a$$

Q. Find the equivalent CFG with no useless variables.

$$G: S \rightarrow AB / Ae$$

$$A \rightarrow aAb / bAa / a$$

$$B \rightarrow bBA / aAB / AB$$

$$C \rightarrow aC / a / aDb$$

$$D \rightarrow bD / aC$$

generating variable should eventually
at some point produce ONLY TERMINALS.
this doesn't happen for C and D so they are
non generating $\rightarrow$ useless productions

Ans. C and D are not reachable, so they are ~~useless~~

Here, all the productions are ^{reachable} generating but C and D are non-generating. Hence, C and D are useless productions that are removed.

$$G': S \rightarrow AB \&$$

$$A \rightarrow aAb / bA / a$$

$$B \rightarrow bBA / aAB / AB$$

Q. Eliminate all useless productions:

$$S \rightarrow aS / AB$$

$$A \rightarrow bA$$

$$B \rightarrow AA$$

Ans. All ~~are~~ are non-generating even though they are reachable.

The language generated by this grammar G, $L(G) = \{\}$

Hence, all are ~~are~~ useless productions and this language generates NULL LANGUAGE (it does not generate even ϵ).

Q. Convert given grammar G to CNF:

$$G: S \rightarrow aA / a / B / \epsilon$$

$$A \rightarrow aB / \epsilon$$

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$$B \rightarrow Aa$$

$$C \rightarrow cCD$$

$$D \rightarrow ddd$$

Ans. Remove all useless productions:

C is reachable, but non-generating production. Hence, remove C.

If C is removed, D becomes unreachable. Hence, D is useless production.

The resultant grammar after removing C and D:

$$S \rightarrow a | aA | B$$

$$A \rightarrow aB | \epsilon$$

$$B \rightarrow Aa$$

Step 1: Remove nullable variables:

$$NV = \{A\}$$

$$S \rightarrow a | aA | B$$

$$A \rightarrow aB$$

$$B \rightarrow Aa | a$$

Step 2: Remove unit productions:

$$S \rightarrow a | B | \cancel{aA} | Aa$$

$$\cancel{A} \rightarrow \cancel{a}$$

$$A \rightarrow \cancel{a}B | Aa | a$$

$$B \rightarrow \cancel{A} \cancel{a} | aB$$

Step 3: Convert to CNF:

$$S \rightarrow a | \cancel{B} | \cancel{X_a}A | AX_a$$

$$A \rightarrow AX_a | a$$

$$B \rightarrow \cancel{a} \cancel{X_a}B$$

$$X_a \rightarrow a$$

Q. Convert G to CNF:

$$G: S \rightarrow aA | aBB$$

$$A \rightarrow aA | \epsilon$$

$$B \rightarrow bB | bC$$

$$C \rightarrow B$$

Ans. Here, B and C are reachable, but non-generating. Hence, useless productions. The resultant grammar:

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$$S \rightarrow aA$$

$$A \rightarrow aaA | \epsilon$$

Step 1: Remove NV

$$NV = \{A\}$$

$$S \rightarrow aA | a$$

$$A \rightarrow aaA | aa$$

Step 2: Remove unit production

No unit production so skip this step.

Step 3: Convert to CNF

$$S \rightarrow X_a A | a$$

$$X_a \rightarrow a$$

$$A \rightarrow X_a T_1 | X_a X_a$$

$$T_1 \rightarrow X_a A$$

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Q. Find a CFG with no ϵ -productions that generate the same language except possibly for ϵ :

Ans.

1) $G:$

$$S \rightarrow AB | \epsilon$$

$$A \rightarrow aASb | a$$

$$B \rightarrow bS$$

Ans. $NV = \{S\}$

$G': S \rightarrow AB$

$$A \rightarrow aASb | a | aAb$$

$$B \rightarrow bS | b$$

NOTE: Strictly removing the ϵ -productions gives $L(G') = L(G) - \{\epsilon\}$.

Q. If ϵ should also be in language, then:

$G':$

$$S \rightarrow AB | \epsilon$$

$$A \rightarrow aASb | a | aAb$$

$$B \rightarrow bS | b$$

2) $G:$

$$S \rightarrow AB | ABC$$

$$A \rightarrow BA | BC | \epsilon | a$$

$$B \rightarrow AC | CB | \epsilon | b$$

$$C \rightarrow BC | AB | A | c$$

Ans. $NV = \{S, A, B, C\}$

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$$G: S \rightarrow AB | ABC | A | B | AB | AC | BC | C$$

$$A \rightarrow BA | BC | a | A | B | C$$

$$B \rightarrow AC | CB | b | A | C | B$$

$$C \rightarrow BC | AB | A | a | B | C$$

$$\text{NOTE: } L(G) = L(G) - \{e\}$$

Q. Find the CFG G' with no ϵ -productions and no unit productions that generate the language $L(G) - \{e\}$:

$$\begin{aligned} 1) \quad G: & S \rightarrow aSa | bSb | \epsilon \\ & A \rightarrow aBb | bBa \\ & B \rightarrow aB | bB | \epsilon \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Even length palindrome}$$

Ans. A, B are non-reachable $\Rightarrow$ Useless variables

$$NV = \{S\}$$

$$G': S \rightarrow aSa | bSb | aa | bb \quad \rightarrow \text{No unit production}$$

$$\text{CNF: } S \rightarrow X_a T_1 | X_b T_2 | X_a X_a | X_b X_b$$

$$X_a \rightarrow a$$

$$T_1 \rightarrow S X_a$$

$$X_b \rightarrow b$$

$$T_2 \rightarrow S X_b$$

$$\begin{aligned} 2) \quad G: & S \rightarrow A | B | C \\ & A \rightarrow aAa | B \\ & B \rightarrow bB | bb \\ & C \rightarrow aCaa | D \\ & D \rightarrow bad | abD | aa \end{aligned}$$

Ans. No useless variables present.

No NV

$$G': S \rightarrow aAa | bB | bb | aCaa | bad | abD | aa$$

$$A \rightarrow aAa | bB | bb$$

$$B \rightarrow bB | bb$$

$$C \rightarrow aCaa | bad | abD | aa$$

$$D \rightarrow bad | abD | aa$$

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Q. Given the CFG, find an equivalent CFG with no useless variables:

1) $G: S \rightarrow ABC \mid BaB$
 $A \rightarrow aA \mid BaC \mid aaa$
 $B \rightarrow bBb \mid a$
 $C \rightarrow CA \mid AC$

Ans. C is non-generating $\Rightarrow$ Useless variables

$G': S \rightarrow BaB$
 $A \rightarrow aA \mid aaa$
 $B \rightarrow bBb \mid a$

A is unreachable $\Rightarrow$ Useless variable

$G': S \rightarrow BaB$
 $B \rightarrow bBb \mid a$

2) $G: S \rightarrow AB \mid AC$
 $A \rightarrow aAb \mid bAa \mid a$
 $B \rightarrow bA \mid aaB \mid AB$
 $C \rightarrow abCa \mid aDb$
 $D \rightarrow bD \mid aC$

Ans. C, D are reachable but non-generating $\Rightarrow$ Useless variables

$G: S \rightarrow AB$
 $A \rightarrow aAb \mid bAa \mid a$
 $B \rightarrow bA \mid aaB \mid AB$

Q. Given the CFG, find a CFG G' in CNF generating $L(G) - \{\epsilon\}$:

1) $G: S \rightarrow SS \mid (S) \mid \epsilon$

Ans. $NV = \{S\}$ $S \rightarrow SS \mid S \mid (S) \mid ($

Eliminate unit production: $S \rightarrow SS \mid (S) \mid ($

Converting to CNF: $S \rightarrow SS$ $X_1 \rightarrow)$
 $S \rightarrow X_1 T_1$ $S \rightarrow X_1 X_2$
 $T \rightarrow SX_2$
 $X_1 \rightarrow ($

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(GNF) Greibach Normal Form - In Peter Linz

The CFG to be in GNF: $\rightarrow S_1, S_2$ are mandatory.
 $\rightarrow S_3$ is optional, but preferred.

A grammar $G = (V, T, P, S)$ is said to be in GNF if all the productions are of the form:

$$A \rightarrow a\alpha \text{ where } A \in V, a \in T, \alpha \in V^*$$

NOTE: The body of the production starts with the terminal followed by 0 or more ~~non~~ variables.

Eg: $S \rightarrow aS \mid bSS \mid a$ ✓ IN GNF
 $S \rightarrow SS$ X NOT IN GNF

Q. Convert the following to GNF:

$$S \rightarrow aSb \mid bSa \mid a \mid b$$

Ans. GNF G' : $S \rightarrow aS_xb \mid bS_xa \mid a \mid b$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

Q. Convert to GNF: $S \rightarrow aSb \mid bSa \mid a \mid b \mid \epsilon$

Ans. $NV = \{S\}$

$$S \rightarrow aSb \mid bSa \mid a \mid b \mid ab \mid ba$$

GNF G' : $S \rightarrow aS_xb \mid bS_xa \mid a \mid b \mid aX_b \mid bX_a$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

CNF: $S \rightarrow XT_1 \mid X_bT_2 \mid a \mid b \mid X_aX_b \mid X_bX_a$

$$T_1 \rightarrow SX_b$$

$$X_b \rightarrow b$$

$$X_a \rightarrow a$$

$$T_2 \rightarrow SX_a$$

P.T.O.

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Left Recursion

A production of the form:

$$A \rightarrow A\alpha \quad \text{where } A \in V \\ \alpha \in (V \cup T)^*$$

is called left recursion.

Eg: Consider the unambiguous validation arithmetic expression grammar:

$$E \rightarrow E+T \mid E-T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid a$$

Here, $E \rightarrow E+T$

$$E \rightarrow E-T$$

$$T \rightarrow T * F$$

$$T \rightarrow T / F$$

are left recursion while others are not.

Removal of Left Recursion

Let $A \rightarrow \underbrace{A\alpha_1 \mid A\alpha_2 \mid \dots \mid A\alpha_m}_{\text{Left Recursion}} \mid \underbrace{\beta_1 \mid \beta_2 \mid \beta_3 \mid \dots \mid \beta_m}_{\text{Non-Left Recursion}}$, then

the equivalent grammar after removing left recursion is:

$$A \rightarrow \beta_1 A' \mid \beta_2 A' \mid \dots \mid \beta_m A'$$

$$A' \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \dots \mid \alpha_m A' \mid \epsilon$$

Eg: $G: A \rightarrow Aa \mid b \Rightarrow L = \{b, ba, baaa, \dots\}$ — Has left recursion

Ans. $\alpha_1 \rightarrow a$

$$\beta_1 \rightarrow b$$

$$G': A \rightarrow bA' \quad \} \Rightarrow L = \{b, ba, \dots\}$$

$$A' \rightarrow aA' \mid \epsilon$$

Q. Find the non-left recursion for the grammar:

$$G: E \rightarrow E+T \mid E-T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid a$$

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Ans.

$$E \rightarrow E+T \mid E-T \mid T$$

$$\alpha_1 = +T$$

$$\alpha_2 = -T$$

$$\beta_1 = T$$

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid -TE' \mid \epsilon$$

$$T \rightarrow T*F \mid T/F \mid F$$

$$\alpha_1 = *F$$

$$\alpha_2 = /F$$

$$\beta_1 = F$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid /FT' \mid \epsilon$$

$$F \rightarrow (E) \mid a$$

$$G': E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid -TE' \mid \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid /FT' \mid \epsilon$$

$$F \rightarrow (E) \mid a$$

$$S_1: NV = \{E', T',$$

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Convert the following grammar

Dependency Graph

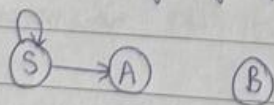
Eg: Let grammar be:

$$G: S \rightarrow aS \mid A$$

$$A \rightarrow aa$$

$$B \rightarrow aa$$

then dependency graph is



In this, B is unreachable. So B is useless.

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Definition:

Dependency Graphs are a way of visualising complex relationships and are found in various applications.

For CFGs, a dependency graph has its vertices labelled with variables, with an edge between the vertices, say C and D if and only if there is a production of the form $C \rightarrow \alpha D \gamma$.

A variable is ~~used~~ to useful if and only if there is a path from the vertex, say C, to the vertex to, say D. In the example, B is unreachable. Hence, B is useless.

Q. Find the CFG without ϵ -production equivalent to grammar:

$G_1: S \rightarrow ABaC$

$A \rightarrow BC$

$B \rightarrow b | \epsilon$

$C \rightarrow D | \epsilon$

$D \rightarrow d$

Ans. No useless variables (All are reachable and generating) at this step.

$NV = \{B, C, A\}$

$G'_1: S \rightarrow ABaC | BaC | AaC | ABa | aC | Ba | Aa | a$

$A \rightarrow BC | C | B$

$B \rightarrow b$

$C \rightarrow D$

$D \rightarrow d$

NOTE: $L(G'_1) = L(G_1) - \{\epsilon\} \rightarrow$ Not mentioned as $\epsilon \notin L(G_1)$ as $\epsilon \notin L(G_1)$

Q. For the given grammar, remove all unit productions:

$G_1: S \rightarrow Aa | B$

$B \rightarrow A | bb$

$A \rightarrow a | bc | B$

Ans. $G'_1: S \rightarrow Aa | bb | a | bc$

$B \rightarrow bb | a | bc$

$A \rightarrow a | bc | bb$

B is unreachable, so B is useless.

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So, $G: S \rightarrow Aa \mid a \mid bc \mid bb$
 $A \rightarrow a \mid bc \mid bb$

Q. Show that the two grammars are equivalent:

$G_1: S \rightarrow abAB \mid ba$
 $A \rightarrow aaa$
 $B \rightarrow aA \mid bb$

$G_2: S \rightarrow abAaA \mid abAbb \mid ba$
 $A \rightarrow aaa$

Ans. In G_2 In G_1 , replace B with respective productions:

$S \rightarrow abAaA \mid abAbb \mid ba$

$A \rightarrow aaa$

We get G_2 .

So, G_1 and G_2 are equivalent.

Q. Eliminate useless productions from:

$G: S \rightarrow aS \mid AB$
 $A \rightarrow bA$
 $B \rightarrow AA$

What language does this G generate.

Ans. S, A, B are useless as they are reachable, but non-generating.
 $L(G) = \{ \}$

Q. Eliminate useless productions:

$G: S \rightarrow a \mid aA \mid B \mid C$
 $A \rightarrow aB \mid E$
 $B \rightarrow Aa$
 $C \rightarrow aCD$
 $D \rightarrow ddd$

Ans. C is useless $\Rightarrow$ Non-generating
D is useless $\Rightarrow$ Non-reachable

$G': S \rightarrow a \mid aA \mid B$
 $A \rightarrow aB \mid E$
 $B \rightarrow Aa$

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Q. Eliminate ϵ -production

$$G: S \rightarrow AaB \mid aaB$$

$$A \rightarrow \epsilon$$

$$B \rightarrow bA \mid \epsilon$$

Ans. $NV = \{A, B\}$ $S \rightarrow A$

$$S \rightarrow AaB \mid aB \mid Aa \mid a$$

A

$$B \rightarrow bA \mid b$$

$$S \rightarrow aaB \mid aa$$

$$G': S \rightarrow aB \mid a \mid aaB \mid aa$$

$$B \rightarrow b$$

Q. Remove all unit productions, useless productions and ϵ -productions from:

$$G: S \rightarrow aA \mid aBB$$

$$A \rightarrow aaA \mid \epsilon$$

$$B \rightarrow bB \mid bC$$

$$C \rightarrow B$$

Ans. B, C are useless $\Rightarrow$ [Reachable but non-generating]

$$S \rightarrow aA$$

$$A \rightarrow aaA \mid \epsilon$$

Remove ϵ -production:

$$NV = \{A\}$$

$$S \rightarrow aA \mid a$$

$$A \rightarrow aaA \mid aa$$

No unit production.

Q. Convert the grammar to CNF:

$$G: S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$

Ans. No useless variable, No ϵ -production, No unit production

$$S \rightarrow AT_1$$

$$A \rightarrow X_a T_2$$

$$B \rightarrow AX_c$$

$$T_1 \rightarrow BX_a$$

$$T_2 \rightarrow X_a X_b$$

$$X_c \rightarrow c$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

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Q. Convert the grammar to GNF:

G: $S \rightarrow AB$
 $A \rightarrow aA \mid bB \mid b$
 $B \rightarrow b$

Ans. GNF: $S \rightarrow AB$
 $S \rightarrow aAB \mid bBB \mid bB$
 $A \rightarrow aA \mid bB \mid b$
 $B \rightarrow b$

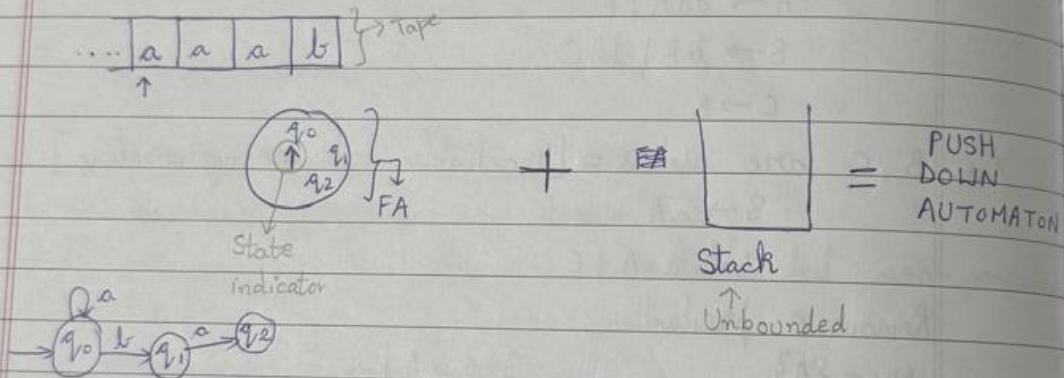
Q. Convert the grammar to GNF:

G: $S \rightarrow aSb \mid aa$

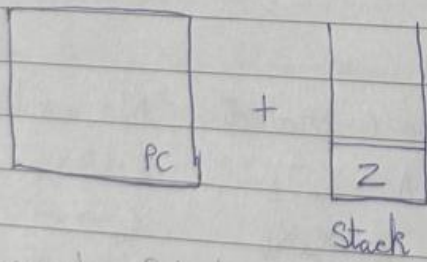
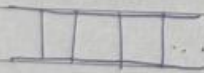
Ans. GNF: $S \rightarrow aX_bSX_b \mid aX_a$
 $X_a \rightarrow a$
 $X_b \rightarrow b$

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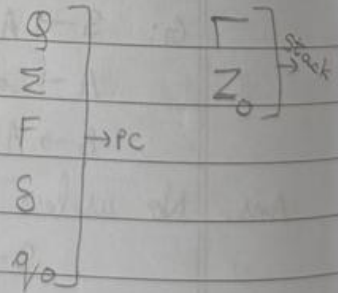
Notes:



General Template of Push Down Automaton



7 tuples:



In general, push down automaton means non-deterministic push down automaton (NPDA)

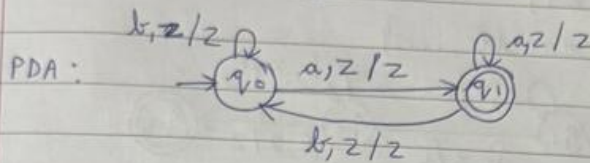
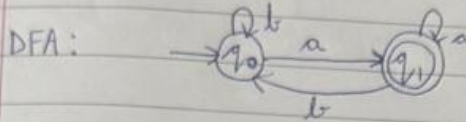
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$L = \{w \mid w \text{ ends with } a\}$
 $\Sigma = \{a, b\}$

Valid $L = \{a, ba, aa, aaa, lbaa, \dots\}$



Push Down Automaton

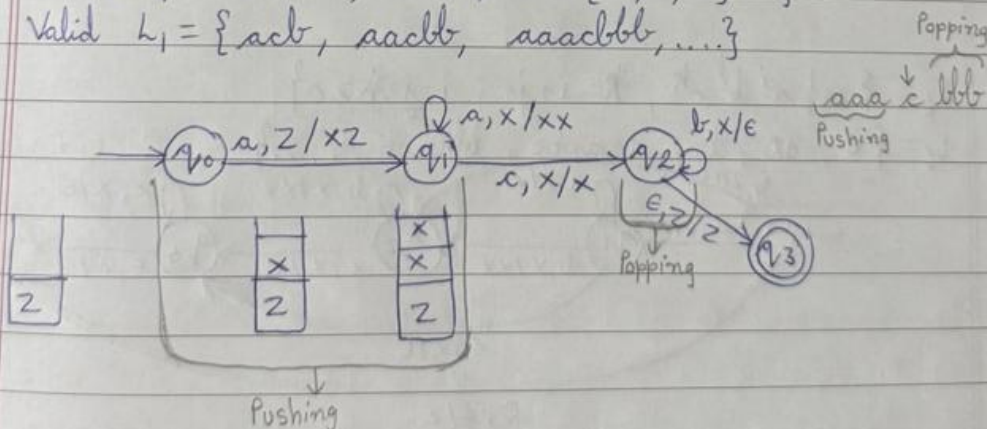
The following transitions are valid in a PDA:

- $x \rightarrow yx$ Push
- $x \rightarrow \epsilon$ Pop
- $x \rightarrow x$ Do Nothing

Q. Write a PDA for the language:

i) $L_1 = \{w \mid a^n c b^m, n \geq 1, w \in \{a, b, c\}^*\}$

Ans. Valid $L_1 = \{acb, aacbb, aaacbbb, \dots\}$



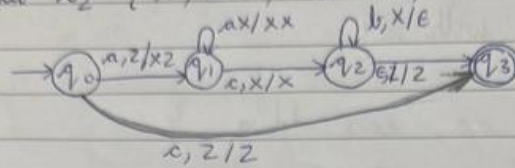
ii) $L_2 = \{w \mid a^n c b^n, n \geq 0, w \in \{a, b, c\}^*\}$

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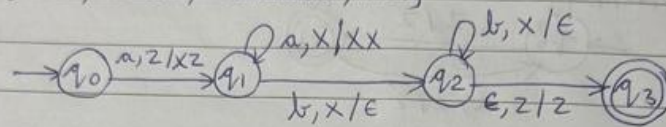
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Ans. Valid $L_2 = \{ \epsilon, ab, aacb, aaacbb, \dots \}$



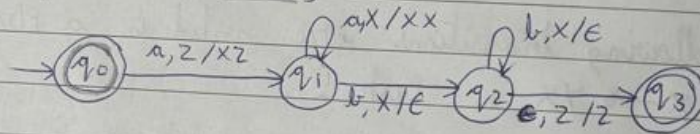
iii) $L_3 = \{ w \mid a^n b^n, n \geq 1 \}$

Ans. $L_3 = \{ ab, aabb, aaabbb, \dots \}$



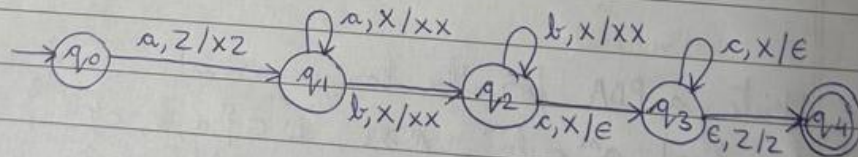
iv) $L_4 = \{ w \mid a^n b^n, n \geq 0 \}$

Ans.



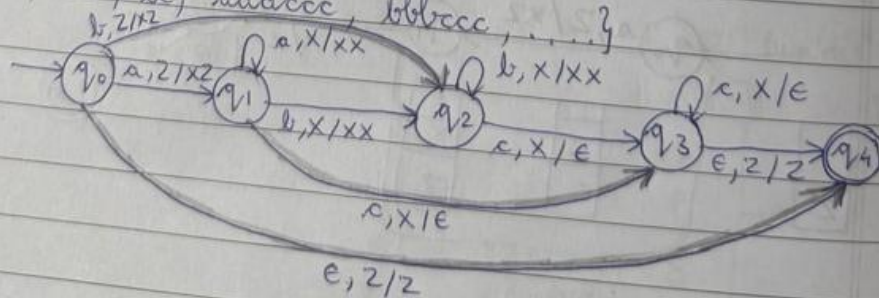
v) $L_5 = \{ w \mid a^i b^j c^k, k = i+j, i, j, k \geq 1 \}$

Ans. $L_5 = \{ abcc, aabccc, abbccc, \dots \}$



vi) $L_6 = \{ w \mid a^i b^j c^k, k = i+j, i, j, k \geq 0 \}$

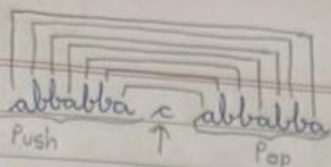
Ans. $L_6 = \{ \epsilon, ac, bc, aaaccc, bbbccc, \dots \}$



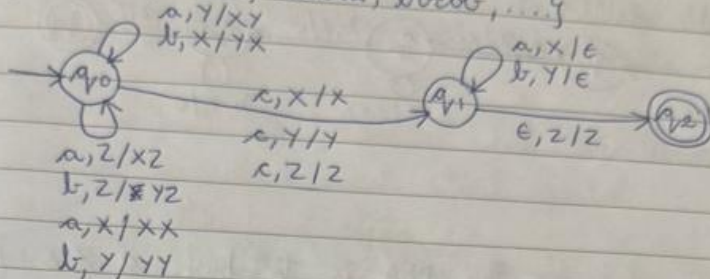
v) $L_7 = \{ w \in w^R \mid w^R \text{ is reverse of } w \}, \Sigma = \{a, b\}$

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Ans.



$$L = \{ \epsilon, aca, bcb, ababa, bbb, \dots \}$$



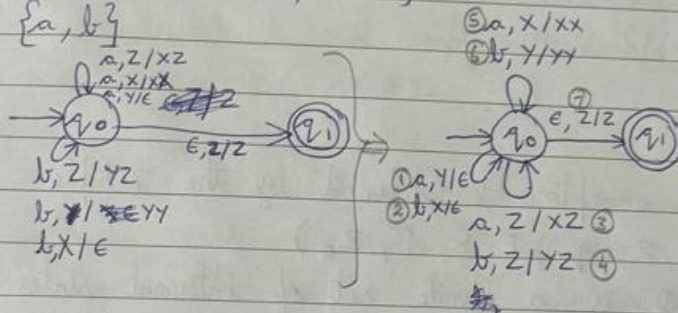
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vi) $L_g = \{ w \mid n_a(w) = n_b(w) \}$

→ Cannot write DPDA for this

Ans. $L = \{ \epsilon, ab, ba, aabb, bbaa, abab, baba, abba, baab, a^n b^n, b^n a^n, \dots \}$

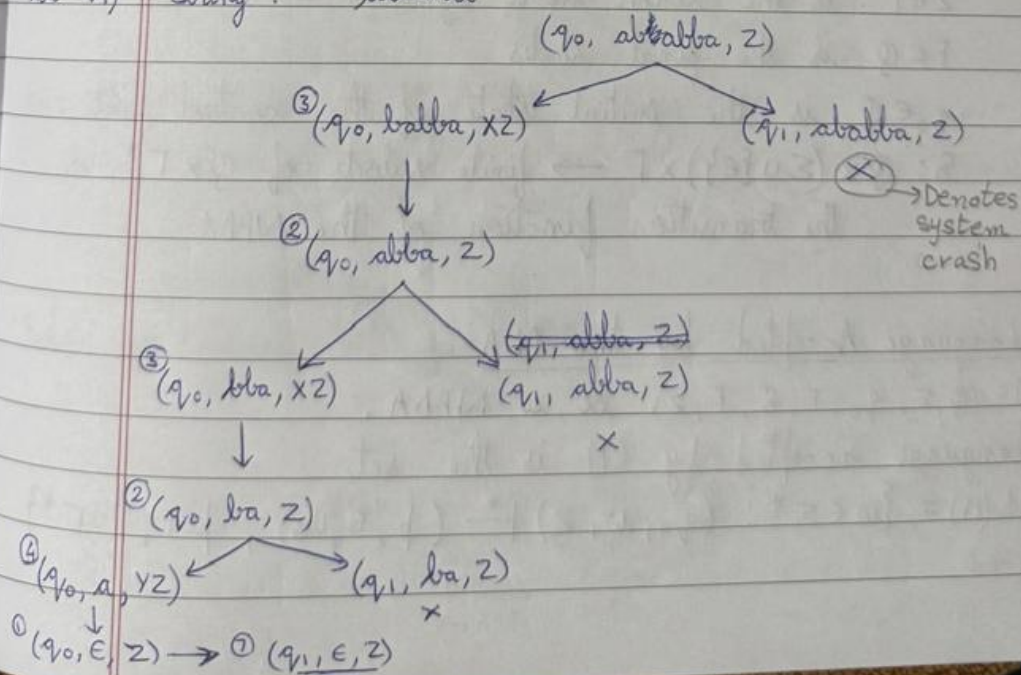
$$\Sigma = \{a, b\}$$



Note: In this example, there is no segregation of push and pop operation.

Instantaneous Description (ID) → 21

For vi) String: ababba



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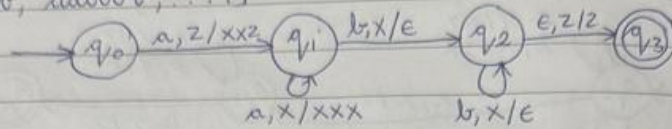
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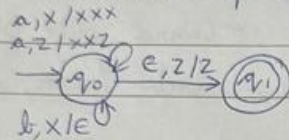
Q. Draw the PDA for the language

$$L = \{a^n b^{2n}, n \geq 1\}$$

Ans. $L = \{ab, aabb, \dots\}$



Note: The idea for this PDA is to push 2 X's for every 'a' scanned. Pop one X for every 'b' scanned.



This is for $L = \{a^n b^{2n}, n \geq 0\}$ as includes $\epsilon \in L$

NPDA

A NPDA acceptor is defined by the 7-tuple

$$M = (Q, \Sigma, q_0, F, \delta, \Gamma, z_0)$$

where Q is a finite set of internal states of the control unit $\rightarrow$ Automaton

Σ is the input alphabet

Γ is a finite set of symbols called the stack alphabet

$z_0 \in \Gamma$ is the stack start symbol.

$F \subseteq Q$ is the final states

$q_0 \in Q$ is the initial state of the control unit

$\delta: Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \rightarrow \text{finite subsets of } Q \times \Gamma^*$ is the transition function of the NPDA

The Language Accepted by the PDA

let $M = (Q, \Sigma, q_0, F, \delta, \Gamma, z_0)$ be a NPDA.

The language accepted by M is the set

$$L(M) = \{w \in \Sigma^* : (q_0, w, z_0) \vdash^* (p, \epsilon, u), p \in F, u \in \Gamma^*\}$$

$\vdash$ $\rightarrow$ Derivation symbol

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DPDA

A PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, Z, F)$ is said to be deterministic if it is an automaton subject to the restrictions where Q is a finite set of internal states of the control unit
 Σ is the input alphabet
 Γ is a finite set of symbols called the stack alphabet
 $Z \in \Gamma$ is the start stack start symbol
 $F \subseteq Q$ is the final states
 $q_0 \in Q$ is the initial state of the control unit
 $\delta: Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \rightarrow \text{finite subsets of } Q \times \Gamma^*$ is transition function of PDA, subject to restrictions that

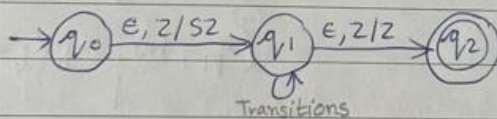
$\forall q \in Q, a \in \Sigma \cup \{\epsilon\}, b \in \Gamma$:

- 1) $\delta(q, a, b)$ contains at most one element
- 2) If $\delta(q, \epsilon, b)$ is not empty, then $\delta(q, a, b)$ must be empty for every $a \in \Sigma$.

To understand: NOTE: For any given input symbol and any stack top atmost one move can be made.
 When a ϵ move is possible for some configuration, then no input consuming alternative is available.

Construction of PDA for the given CFG

Template:



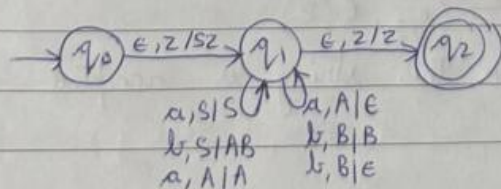
Ex: Q. Construct PDA for the given CFG:

$S \rightarrow aS \mid bAB$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

Soln: 1) Convert the given CFG to GNF



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8. Convert CFG to PDA:

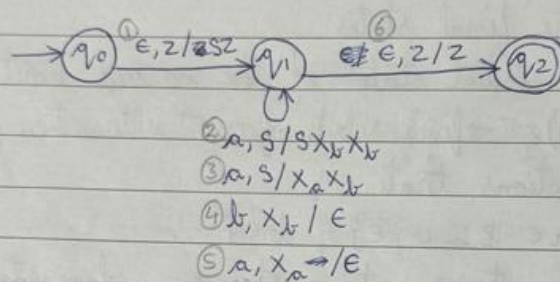
i) $S \rightarrow asbb \mid aab$

Ans. GNF: $S \rightarrow aSx_bx_b \mid aX_aX_b$

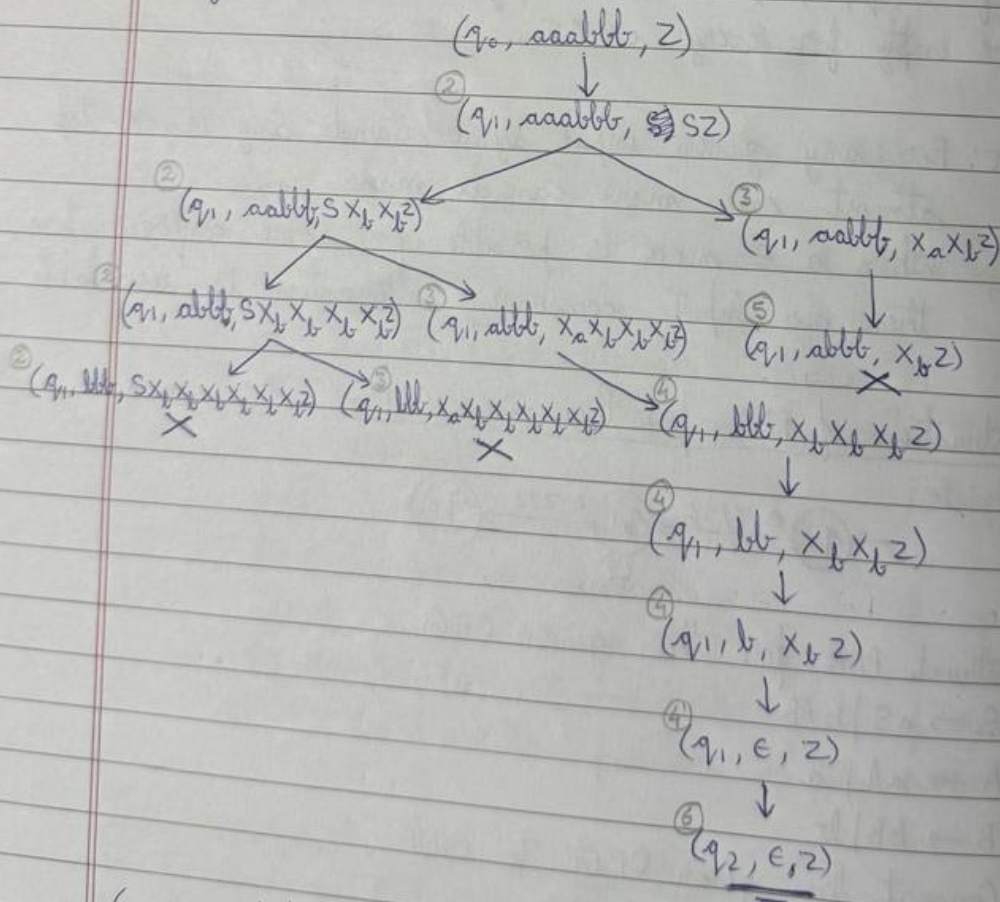
$X_a \rightarrow a$

$X_b \rightarrow b$

PDA:



String: aaabbb



$(q_0, aaabbb, z) \vdash^* (q_2, \epsilon, z)$

Since $q_2 \in F$ so this string is accepted

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Pumping lemma for Context Free Languages

THEOREM:

Let 'L' be an infinite context free language. Then there exists some positive integer 'm' such that any $w \in L$ with $|w| \geq m$ can be decomposed as $w = uvxyz$ with $|vxy| \leq m$ and $|vxy| \geq 1$

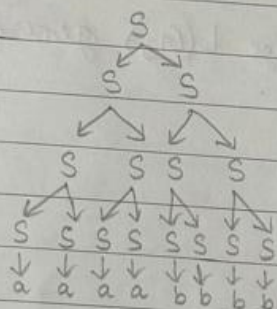
such that $uv^i xy^i z \in L$ for all $i = 0, 1, 2, \dots$

This is known as pumping lemma for context free language.

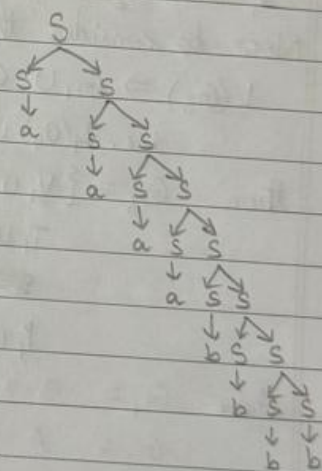
Let CFG $S \rightarrow SS \mid a \mid b$ - Accepts everything except ϵ

String: $aaaabbbb$

Derivation Trees:

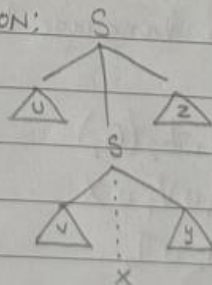


Balanced



Skewed

PUMPING LEMMA REPRESENTATION:



Q. Prove L is not Context free language: $L = \{a^m b^m c^m, m > 0\}$

Ans. String $aaabcc \in L$.

$|wxyz| \leq m$

$u = a$
 $x = \epsilon$
 $y = a$

$\Rightarrow uv^i xy^i z$ with $u = \epsilon, y = abcc$.

$i=2: uav^2abcc \Rightarrow aaabcc \notin L$
 $\therefore$ Not CFL.

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Closure Properties of Decision Algorithms for Context Free languageTHEOREM:

The family of context free languages is closed under union, concatenation and star closure (Kleene star).

NOTE: The family of CFL is not closed under intersection and complementation.

PROOF:

Let L_1 and L_2 be 2 CFLs generated by the grammar CFGs $G_1 = (V_1, T_1, S_1, P_1)$

$G_2 = (V_2, T_2, S_2, P_2)$ respectively.

We can assume that without the loss of Generality that V_1 and V_2 are disjoint.

Now to consider the language $L(G_3)$ generated by the grammar

$$L(G_3) \Rightarrow G_1 \cup G_2$$

$$\text{i.e. } L(G_1) \cup L(G_2)$$

then $G_3 = \{V_1 \cup V_2 \cup X,$

$$T_1 \cup T_2,$$

$$X,$$

$$P_1 \cup P_2 \cup \{X \rightarrow S | A\}\}. \text{ Hence, proved.}$$

where $G_1 = S \rightarrow SS | a | b$

$$G_2 = A \rightarrow aA | a$$

$$G_3 = X \rightarrow S | A \text{ \& } S \& A \text{ from } G_1 \& G_2 \text{ respectively}$$

Now consider the language $L(G_4)$ generated by grammar

$$L(G_4) \Rightarrow G_1 G_2$$

$$\text{i.e. } L(G_1) \cdot L(G_2)$$

then $G_4 = \{V_1 \cup V_2 \cup X$

$$T_1 \cup T_2$$

$$X$$

$$P_1 \cup P_2 \cup \{X \rightarrow SA\}\}. \text{ Hence, proved.}$$

Now consider, the language $L(G_5)$ generated by grammar

$$L(G_5) = G_1^*$$

then $G_5 = (V_1, T_1, S_1, P_1 \cup \{S_1 \rightarrow S_1 S_1 | \epsilon\})$

$$G_1 = S \rightarrow S | a | b$$

$$G_5 = S_1 \rightarrow S_1 S_1 | \epsilon \text{ \& } S \text{ from } G_1$$